Ishaan Agarwal

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Education

Cornell University, Ann S. Bowers College of Computing and Information Science Ithaca, NY
Master of Professional Studies, Applied Statistics GPA: 3.87/4.0 *May 2026*

Coursework: Data Mining & Machine Learning, Applied Neural Networks, Introduction to Deep Learning, Bayesian Data Analysis, Applied Time Series Analysis, Modern Regression, Categorical Data, Statistical Computing.

Manipal Institute of Technology Karnataka, India
B.Tech, Electronics and Communication Engineering Minor: Data Science *June 2023*

Experience

UBS Bank India
Software Engineer, Data & Search Infrastructure (Internship → Full-Time) *January 2023 – July 2025*

- Prototyped a **RAG pipeline** to auto-tag 10,000+ financial documents using LLM-generated tags from semantically similar document context, reducing manual tagging effort by 60%; **placed Top 5 in UBS AI Quest Competition**.
- Owned 20+ Azure Search indexes powering UBS Neo's cross-asset trading platform serving **1.8M+ users and 200K daily API calls**, with real-time ingestion from Kafka, Tibco, and REST APIs.
- Led end-to-end migration of a business-critical search service to a cloud-native AKS architecture, replacing service-level auth with platform-wide verification; shipped with **zero production incidents**.
- Implemented region-based data segmentation for the FX search pipeline by extending the Azure Search schema and executing a full reindex of **2.4M+ financial instrument records**.
- Migrated core services from 29West to gRPC and to Azure Cache for Redis, enabling AKS deployment and maintaining a real-time data-quality monitoring pipeline across upstream sources.

Research

Embedding-Based Causal Adjustment for Latent Confounders Cornell University
Advisor: Prof. Jelena Bradic Cornell SDS Project Showcase *January – May 2026*

- Investigated whether **LLM-derived text embeddings** of household histories can serve as valid proxies for latent confounders (motivation, labor-market attachment) when estimating the causal effect of job training on earnings.
- Designed a **semi-synthetic benchmark** on 6,107 PSID households with injected latent confounders, spanning linear-to-threshold outcome regimes and constant-to-nonlinear-heterogeneous treatment effects.
- Built and benchmarked **7 embedding pipelines**: TF-IDF+SVD, Sentence-Transformers (MiniLM), trajectory embeddings over panel waves, multi-view fusion (concat/linear/attention-weighted), adversarial overlap-aware projection, and contrastive stability-tuned ensembles.
- Estimated Average Treatment Effects via **Double Machine Learning with AIPW and cross-fitting**; best embedding-based estimator reduced **RMSE by 2–3.4×** and **bias by 55%** versus covariate-only adjustment.
- Applied the framework to real PSID data (waves 2007–2021), finding that job training has **no statistically significant effect on household income**.

Projects

Inductive Bias & Sample Efficiency: CNNs vs Vision Transformers

- Benchmarked a lightweight ResNet against a Minimal ViT on progressive data splits (1%–100%) to quantify how spatial inductive bias trades off against data scale.
- Found the CNN reached **84.55% test accuracy with only 5% of training data**, while the ViT required the full dataset plus augmentation to reach a 95.66% peak, quantifying a ∼20× data-efficiency gap.

High-Dimensional Gene Network Analysis *$p \gg n$ setting*

- Estimated sparse precision matrices from breast cancer expression data by implementing **Constrained L1-Minimization (CLIME)** and **Graphical Lasso** from scratch; reconstructed gene regulatory networks to predict clinical response via LDA, benchmarking convex (CLIME) versus non-convex (SCAD) penalization.

Skills

Languages & Libraries:	Python, R, SQL, PyTorch, Scikit-learn, HuggingFace, Pandas, NumPy, Matplotlib
ML & Statistics:	Causal Inference (DML, AIPW), Time Series Forecasting, Deep Learning, RAG, LLM Embeddings, Computer Vision, High-Dimensional Statistics
Cloud & Infrastructure:	Azure (AKS, Search, Redis), Kafka, Kubernetes, GitLab CI/CD, PostgreSQL, REST APIs
Certifications:	Microsoft Azure AI Fundamentals (AI-900); Deep Learning Specialization (DeepLearning.AI)